(MTH603) (Final Term)

(Lecture #23)

(Lagrange's Interpolation Formula)

$$f(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} f(x_0) + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} f(x_1) + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} f(x_2)$$

(Lectur #24)
(Divided Differences)

=> Zero-order divided differences;

$$Y[x_0] = Y_0$$

=> The First order divided differences;

$$Y[x_0, x_1] = \frac{Y_1 - Y_0}{x_1 - x_0}$$

=> 2nd order divided difference;

$$Y[x_0, x_1, x_2] = \frac{Y[x_1, x_2] - Y[x_0, x_1]}{x_2 - x_0}$$

Example;

Construct the Newton's divided difference table for values of

$X = 1, 2, 3, 4, 5, 6$

and

$f(x) = -3, 0, 15, 48, 105, 192.$

x	f(x)	1st D	2nd D	3rd D
1	-3			
2	0	$0-(-3)=3$		
3	15	$15-0=15$	$\frac{15-3}{3-1}=6$	
4	48	$48-15=33$	$\frac{33-15}{4-2}=9$	$\frac{9-6}{4-1}=1$
5	105	$105-48=57$	$\frac{57-33}{5-3}=12$	$\frac{12-9}{5-2}=1$
6	192	$192-105=87$	$\frac{87-57}{6-4}=15$	$\frac{15-12}{6-3}=1$

Newton's divided difference can also be expressed in term of Forward, backward and Central difference.

$$Y[x_0, x_1] = \frac{\Delta Y_0}{h}, \frac{\nabla Y_0}{h}$$

$$Y[x_0, x_1, x_2] = \frac{\Delta^2 Y_0}{2! h^2}, \frac{\nabla^2 Y^0}{2! h^2}$$

$$y[x_0, x_1] = \frac{\delta y_{1/2}}{h}$$

$$y[x_0, x_1, x_2] = \frac{\delta^2 y_{1/2}}{2! h^2}$$

Newton's Divided difference formula General form;

$$y = f(x) = y_0 + (x - x_0) y[x_0, x_1] + (x - x_0)(x - x_1) y[x_0, x_1, x_2] + \cdots + (x - x_0)(x - x_1) \cdots (x - x_{n+1}) y[x_0, x_1, \cdots x_0]$$

ERROR Term In Interpolation Formular.

$$\Rightarrow \quad \varepsilon(x) = y(x) - P_n(x)$$

$\Rightarrow$ $P_n(x)$ is a polynomial of degree n, its $(n+1)$th derivative is **zero.**

(Lecture # 27)
(Differentiation using Difference operators)

(Forward difference)

$$\Delta = E - 1$$

This would mean that in term of Δ;

$$D = \frac{1}{h}\left[\Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \frac{\Delta^4}{4} + \frac{\Delta^5}{5} - \cdots\right]$$

$$D^2 = \frac{1}{h^2}\left[\Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \frac{\Delta^4}{4} + \cdots\right]^2$$

(Backward difference)

$$D = \frac{1}{h}\left(\nabla - \frac{\nabla^2}{2} + \frac{\nabla^3}{3} + \frac{\nabla^4}{4} + \cdots\right)$$

$$D^2 = \frac{1}{h^2}\left[\nabla - \frac{\nabla^2}{2} + \frac{\nabla^3}{3} + \frac{\nabla^4}{4} + \cdots\right]^2$$

(Lecture # 30)
(Richardson's Extrapolation Method)

$$f(h) = \frac{y(x+h) - y(x-h)}{2h}$$

use the given function at the place of $(x+h)$ & $(x-h)$.

And;

$$F_m\left(\frac{h}{2^n}\right) = \frac{4^m f_{m-1}\left(\frac{h}{2^m}\right) - f_{m-1}\left(\frac{h}{2^{m-1}}\right)}{4^m - 1}$$

=> Trapzoidal Rule;

* 2nd order accurate. $O(h)^2$

Interval $[x_0, x_1]$

=> Simpson's 1/3 Rule;

Interval $[x_0, x_1, x_2]$

(fourth order accurate) => $O(h)^4$

Based on Fiting three point with quadratic.

=> Simpson's 3/8 Rule;

Interval $[x_0, x_1, x_2, x_3]$

Fourth order accurate $(Oh)^4$

Based on Fiting four points by a cubic.

(Numerical Integration & Differentiation)

$$AREA = \int_a^b f(x)dx$$

$$VOLUME = \int_c^d \int_a^b g(x,y)\,dx\,dy$$

Basic defination of integral;

$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k)\Delta x$$

(Formulas)

=> Trapezoidal Rule:-

$$\int_{x_0}^{x_n} f(x)dx = \frac{h}{2}\left[(y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n) + E_n\right]$$

ERROR Term;

$$E_n = -\frac{h^3}{12} y''(\xi_e)$$

$$\left\{ h = \frac{b-a}{n} \right.$$

OR

$$E_n = -\frac{h^2(x_n - x_0)}{12} y''(\xi_e)$$

=> Simpson's 1/3 Rule;

$$\int_{x_0}^{x_n} f(x)dx = \frac{h}{3}(Y_0 + 4(Y_1 + Y_3 + \cdots Y_{n-1}) + 2(Y_2 + Y_4 + \cdots Y_{n-2} + Y_n) + E_n$$

ERROR Term;

$$E_n = -\frac{h^5}{90} Y^{(iv)}(\xi_e)$$

OR;

$$= -\frac{x_{2n} - x_0 h^4}{180} Y^{iv}(\xi_e)$$

=> Simpson's 3/8 Rule;

$$\int_{x_0}^{x_n} f(x)dx = \frac{3h}{8}\left(y_0 + 3y_1 + 3y_2 - \cdots + 2y_{n-3}\right)$$

$$\int_{x_0}^{x_n} f(x)dx = \frac{3h}{8}\left(y_0 + 3y_1 + 3y_2 + 2y_3 + 3y_4 + 3y_5 + 2y_6 + \cdots + 2(y_{n-3}) + 3y_{n-1} + y_n\right)$$

Global ERROR,

$$E = -\frac{x_n - x_0}{80} h^4 y^{iv}$$

=> Percentage of ERROR :-

$$= \frac{|\text{True value}| - |\text{Measured value}|}{|\text{True value}|} \times 100$$

Same for simpson's Rule & Trepezoidal

=> Richardson's extapolation procedure ; to trapezodial Rule

$$E_{a_m}\left(\frac{h}{2^m}\right) = \frac{4^m E_{T(m-1)}\left(\frac{h}{2^m}\right) - E_{T(m-1)}\left(\frac{h}{2^{m-1}}\right)}{4^{m-1}}$$

$m = 1, 2, 3 \cdots$

Points to Remberi;

for h, Take $N = 1$, $\frac{h}{2^3}$, $N = 8$

$\frac{h}{2}$, $N = 2$,

$\frac{h}{2^2}$, $N = 4$,

(ORDinary Differential Equations)

(Taylor Series)

if $y(t)$ is the exact solution we can expand $y(t)$ by Taylor's Series about the point $t=t_0$ and obtain:

$$y(t) = y(t_0) + (t-t_0)y'(t_0) + \frac{(t-t_0)^2}{2!}y''(t_0) + \frac{(t-t_0)^3}{3!}y'''(t_0) + \cdots$$

$$= y_0 + (t-t_0)y_0' + \frac{(t-t_0)^2}{2!}y_0'' + \frac{(t-t_0)^3}{3!}y_0''' + \cdots$$

$\Rightarrow$ we can express any derivative of y in term of $f(t,y)$ and its partial derivates.

(Euler Method)

Formulas used in Solving Examples:

$$y = y_0 + (t-t_0)f(t_0, y_0)$$

$$y_1 = y_0 + (t_1-t_0)f(t_0 y_0)$$

$$y_2 = y_1 + hf(t_1, y_1)$$

$$y_{m+1} = y_m + hf(t_m y_m)$$

$m = 0, 1, 2 \ldots$

$$y_4 = y_3 + hf(y_3, t_3)$$

Example;

$$\frac{dy}{dt} = \frac{y-t}{y+t}$$

Initial Condition $y=1$, at $t=0$

Find y approximately at $x=0.1$ in five steps;

Solution;

$$n = 5$$

Step size $h = \frac{0.1}{5} = 0.02$

$t = 0.02, 0.04, 0.06, 0.08, 0.1$

Thus;

$y_1 = y_0 + hf(t_0, y_0)$, $y_0 = 1$, t_0

$y_1 = 1 + 0.02\left(\frac{1-0}{1+0}\right) = \boxed{1.02}$

$y_2 = y_1 + hf(t_1, y_1)$

$= 1.02 + 0.02 \frac{1.02 - 0.02}{1.02 + 0.02} = 1.0392$

$y_3 = y_2 + hf(t_2, y_2)$

$= 1.0392 + 0.02 \frac{1.0392 - 0.04}{1.0392 + 0.04}$

$= 1.0577.$

(Modified Eules's Method.)

Formulae;

$$y_1 = y_0 + \left[\frac{f(t_0, y_0) + f(t_1, y_1^{(1)})}{2}\right](t_1 - t_0)$$

$$= y_0 + h\left[\frac{f(t_0, y_0) + f(t_1, y_1^{(1)})}{2}\right]$$

$$y_{m+1} = y_m + h\left[\frac{f(t_m, y_m) + f(t_{m+1}, y_{m+1}^{(1)})}{2}\right]$$

$m = 1, 2, 3 \text{ —— } m$

Milne Method:-

Formulas:-

$$Y_4 = y_0 + \frac{4h}{3}(2y_1' + y_2' + 2y_3') + \frac{28}{90}h\Delta^4 y_0'$$

This Milne's predictor Formula.

Milne's predictor-Corrector pair can be written as;

$$P;\ y_{n+1} = y_{n-3} + \frac{4h}{3}(2y'_{n-2} - y'_{n-1} + 2y'_n)$$

$$C;\ y_{n+1} = y_{n-1} + \frac{h}{3}(y'_{n-1} + 4y'_n + y'_{n+1})$$

Magnitude of the truncation ERROR in Corrector Formula;

$$= 1/90 h\Delta^4 y_0'$$

Truncation Error in predictor formula;

$$\frac{28}{90} h\Delta^4 y_0'$$

TE in C-Formula is Less than the TE in P-formula (True)

(Adam's predictor Formula)

$$Y_{n+1} = Y_n + \frac{h}{24}[55y_n' - 59y_{n-1}' + 37y'_{n-2} - 9y'_{n-3}] + \frac{251}{720}h\Delta^4 y'$$

Truncation ERROR $= \left(\frac{251}{720} h\Delta^4 y_n'\right)$

$$y_4 = y_0 + \frac{4h}{3}(2y_1' - y_2' + 2y_3')$$

Milne's predictor - corrector

P: $y_{n+1} = y_{n-3} + \frac{4h}{3}(2y'_{n-2} - y'_{n-1} + 2y'_n)$

C: $y_{n+1} = y_{n-1} + \frac{h}{3}(y'_{n-1} + 4y'_n + y'_{n+1})$

(Predictor - Corrector formulae)

P: $y_{n+1} = y_{n-1} + 2hf(t_n, y_n)$

C: $y_{n+1} = y_n + \frac{h}{2}[(f(t_n, y_n) + f(t_n$

(RK - Method)

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$k_1 = hf(t_1, y_n)$

$k_2 = hf(t + \frac{h}{2}, y_n + \frac{k_1}{2})$

$k_3 = hf(t + \frac{h}{2}, y_n + \frac{k_2}{2})$

$k_4 = hf(t_n + h, y_n + k_3)$

(RUNGE- Kutta Method)
(RK Method)

=> 2nd order Formulae;

$$Y_{n+1} = Y_n + \frac{1}{3}(2k_1 + k_2)$$

$$k_1 = hf(t_n, y_n)$$

$$k_2 = hf\left(t_n + \frac{3h}{2}, y_n + \frac{3k_1}{2}\right)$$

=> 4th order Formulae;

$$Y_{n+1} = Y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = hf(t_n, y_n)$$

$$k_2 = hf\left(t_n + h/2, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf(t_n + h/2, y_n + k_2/2)$$

$$k_4 = hf(t_n + h, y_n, k_3)$$

PREDICTOR - Corrector Method;

P: $y_{n+1} = y_{n+1} + 2hf(t_n + y_n)$

C: $y_{n+1} = y_n + \frac{h}{2}[F(t_n, Y_n) + f(t_{n+1}, y_{n+1})]$

(Two points Formula.)

$$Y_i' = \frac{y(x_i+h) - y(x_i-h)}{2h}$$ ⇒ for Mid

This equation Consitute two-point formula for the first derivative.

forward difference;

$$= \frac{y(x_i+h) - y(x_i)}{h}$$ ⇒ (for last point)

Backward;

$$\frac{y(x_i) - y(x_i-h)}{h}$$ → (for start point)

(Three Points FORMULA)

$$= \frac{y(x_i-h) - 2y(x_i) + y(x_i+h)}{h^2}$$

Simpson's $\frac{3}{8}$;

$$\int_a^b f(x)dx = \frac{3}{8}h[y(a) + 3y_1 + 3y_2 + 2y_3 + 3y_4 + 3y_5 + 2y_6 + \text{---} + 2y_{n-3} + 3y_{n-2} + 3y_{n-1} + y(b)]$$